

Exercise: Rotor Dynamics

2025

1. Consider a rotor blade with uniform mass, with a lag hinge positioned at distance eR from the root. The homogeneous lag equation is given by:

$$\zeta'' + \nu_\zeta^2 \zeta = 0, \quad \text{where } \nu_\zeta^2 = \frac{3e}{2(1-e)} + \frac{k_\zeta}{I_\zeta \Omega^2}$$

- What type of system is this and what system properties does this formulation allow us to determine?
- Find the lag natural frequency.
- Assuming now that the mechanical spring effect is zero, where does the lag hinge need to be placed so that the lag natural frequency is equal to the rotor running frequency Ω ?

(a) **1 DoF system (mass-spring damper)**

ω_n (natural frequency) and ζ (damping) can be determined by
homogenous equation: $x'' + 2\zeta\omega_n x' + \omega_n^2 x = 0$

$$(b) \quad \omega_n = \nu_\zeta \Omega = \sqrt{\frac{3e}{2(1-e)} \Omega^2 + \frac{k_\zeta}{I_\zeta}}$$

$$(c) \quad \text{if } k = 0, \quad \omega_n = \Omega \sqrt{\frac{3e}{2(1-e)}} = \Omega$$

$$\therefore \frac{3e}{2(1-e)} = 1 \quad \therefore 3e = 2 - 2e \quad \therefore e = \frac{2}{5}$$

$$\therefore x = \frac{2}{5}R = 0.4R \text{ (this result seems unrealistic)}$$

2. For a centrally-hinged rigid blade with uniform mass, find the flap damping in hover conditions. Under what conditions is the rotor critically damped?

Assume that the nondimensional flapping moment is given by:

$$\bar{M}_{\beta, \text{hover}} = \frac{\Omega^2 I_b}{\rho c C_{l\alpha} R^4 \Omega^2} \frac{\gamma}{8} \left(\theta - \frac{\dot{\beta}}{\Omega} - \frac{4\lambda_i}{3} \right), \quad \gamma = \frac{\rho c C_{l\alpha} R^4}{I_b}$$

Express the flap by Fourier expansion, only take 1st harmonic term:

$$\beta = \beta_0 + \beta_{1c} \cos \psi + \beta_{1s} \sin \psi$$

$$\therefore \dot{\beta} = -\beta_{1c} \sin \psi + \beta_{1s} \cos \psi$$

$$\therefore \ddot{\beta} = -\beta_{1c} \cos \psi - \beta_{1s} \sin \psi$$

for centrally-hinged rigid motor, $\nu_{\beta} = 1$

$$\ddot{\beta} + \beta = \delta \bar{M}_{\beta} = \frac{\delta}{8} \left(\theta - \frac{\dot{\beta}}{\Omega} - \frac{4}{3} \lambda_i \right) = \frac{\delta}{8} \left(\theta - \frac{\dot{\beta}}{\Omega} - \frac{4}{3} \lambda_i \right)$$

$$\therefore \ddot{\beta} + \frac{\delta}{8} \dot{\beta} + \beta = \frac{\delta}{8} \left(\theta - \frac{4}{3} \lambda_i \right)$$

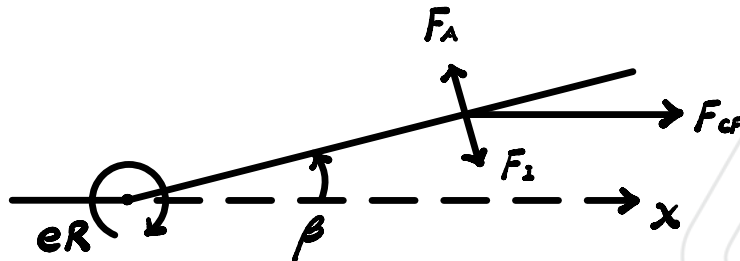
$$\therefore \ddot{\beta} + \frac{\delta}{8} \Omega \dot{\beta} + \Omega^2 \beta = \frac{\delta}{8} \Omega \left(\theta - \frac{4}{3} \lambda_i \right)$$

$$\therefore \omega_n^2 = \Omega^2 \quad \therefore \omega_n = \Omega$$

$$\therefore 2\zeta\omega_n = \frac{\delta}{8} \Omega \quad \therefore \zeta = \frac{\delta}{16}$$

when Lock number, $\delta = 16$, rotor critically damped

3. Derive the forces and moments about a flapping rigid blade, with the flap hinge located at a distance eR from the shaft axis.



for small angle approximation.

$$dF_z = dm(x - eR)\ddot{\beta}$$

$$dF_{cf} = dm\Omega^2 x \quad \therefore dF_{cf,\perp} = dm\Omega^2 x \sin\beta \approx dm\Omega^2 \beta$$

$$dF_A = L' \cos\beta \approx L' = \frac{1}{2}\rho U_T^2 c C_{l\alpha}(\theta - \phi)$$

$$\text{where } \phi = \frac{U_p}{U_T}, \quad U_T = \Omega x, \quad U_p = \dot{\beta}x + v_i$$

$$\therefore dF_A \approx \frac{1}{2}\rho c C_{l\alpha}(U_T^2\theta - U_p U_T)$$

$$= \frac{1}{2}\rho c C_{l\alpha}\Omega^2(\theta x^2 - \frac{\dot{\beta}}{\Omega}x^2 - \frac{v_i}{\Omega}x)$$

Therefore, for small hinge offset, e

$$I_b = \int_0^R mx^2 dx \approx \int_{eR}^R m(x - eR)^2 dx$$

$$\therefore M_z = \int_{eR}^R dF_z(x - eR) dx = I_b \ddot{\beta} = I_b \Omega^2 \beta^{**}$$

$$M_{cf} = \int_{eR}^R m \Omega^2 \beta (x - eR) x \, dx$$

$$= \Omega^2 \beta \int_{eR}^R m (x - eR) x \, dx$$

$$= \Omega^2 \beta \int_{eR}^R m (x - eR) (x - eR + eR) \, dx$$

$$= \Omega^2 \beta \left(\int_{eR}^R m (x - eR)^2 \, dx + \int_{eR}^R m (x - eR) \, dx \right)$$

$$= I_b \Omega^2 \beta \left(1 + \frac{eR \int_{eR}^R m (x - eR) \, dx}{I_b} \right)$$

$$= I_b \Omega^2 v_{\theta^2} \beta$$

$$M_A = \int_{eR}^R L' (x - eR) \, dx$$

$$\therefore \ddot{\beta} + v_{\theta^2} \beta = \frac{1}{I_b \Omega^2} \int_{eR}^R L' (x - eR) \, dx$$

4. Starting from the nondimensional flapping equation in forward flight conditions, for a rigid centrally-hinged blade, determine the coning angle and explain its significance.

$$\beta^{**} + \beta = \gamma \theta \left(\frac{1}{8} + \frac{\mu}{3} \sin \psi + \frac{\mu^2}{4} \sin^2 \psi \right) - \gamma \lambda \left(\frac{1}{6} + \frac{\mu}{4} \sin \psi \right) - \gamma \beta^* \left(\frac{1}{8} + \frac{\mu}{6} \sin \psi \right) - \gamma \beta \mu \cos \psi \left(\frac{1}{6} + \frac{\mu}{4} \sin \psi \right)$$

Express flapping angle by Fourier expansion, take 1st harmonic term:

$$\beta = \beta_0 + \beta_{1c} \cos \psi + \beta_{1s} \sin \psi$$

$$\therefore \dot{\beta} = -\beta_{1c} \sin \psi + \beta_{1s} \cos \psi$$

$$\therefore \beta^{**} = -\beta_{1c} \cos \psi - \beta_{1s} \sin \psi$$

same for pitching angle, $\theta = \theta_0 + \theta_{1c} \cos \psi + \theta_{1s} \sin \psi$

Find coning angle, β_0 , using harmonic matching:

$$\beta_0 = \delta (\theta_0 + \theta_{1c} \cos \psi + \theta_{1s} \sin \psi) \left(\frac{1}{8} + \frac{\mu}{3} \sin \psi + \frac{\mu^2}{4} \sin^2 \psi \right) - \frac{1}{6} \delta \lambda$$

$$\beta_0 = \delta \left[\theta_0 \left(\frac{1}{8} + \frac{\mu^2}{4} \sin^2 \psi \right) + \theta_{1c} \left(\frac{\mu^2}{4} \sin^2 \psi \cos \psi \right) + \theta_{1s} \left(\frac{\mu}{3} \sin^2 \psi + \frac{\mu^2}{4} \sin^2 \psi \right) - \frac{\lambda}{6} \right]$$

$$\text{where } \sin^2 \psi = \frac{1}{2} [1 - \cos(2\psi)]$$

$$\sin^2 \psi \cos \psi = \frac{1}{2} \cos \psi - \frac{1}{2} \cos(2\psi) \cos \psi = \frac{1}{2} \left\{ \cos \psi - \frac{1}{2} [\cos \psi + \cos(3\psi)] \right\}$$

$$\sin^2 \psi = \frac{1}{2} \sin \psi - \frac{1}{2} \cos(2\psi) \sin \psi = \frac{1}{2} \left\{ \sin \psi - \frac{1}{2} [-\sin \psi + \sin(3\psi)] \right\}$$

$$\therefore \beta_0 = \delta \left[\theta_0 \left(\frac{1}{8} + \frac{\mu^2}{8} \right) + \theta_{1c} \left(\frac{\mu}{6} \right) - \frac{\lambda}{6} \right]$$

coning angle is the angle related to hub plane, which represent the equilibrium flapping angle when aerodynamic and centrifugal forces balanced

5. Given the flapping equation in forward flight, for a rigid, untwisted, centrally hinged blade, how would you determine the flap response to lateral cyclic input?

$$\ddot{\beta} + \beta = \gamma\theta \left(\frac{1}{8} + \frac{\mu}{3} \sin \psi + \frac{\mu^2}{4} \sin^2 \psi \right) - \gamma\lambda \left(\frac{1}{6} + \frac{\mu}{4} \sin \psi \right) - \gamma\beta^* \left(\frac{1}{8} + \frac{\mu}{6} \sin \psi \right) - \gamma\beta\mu \cos \psi \left(\frac{1}{6} + \frac{\mu}{4} \sin \psi \right)$$

Assuming an advance ratio of 0.4 and a coning angle of 2° , what is the effect of a 5° lateral cyclic input on the tip path plane? Use the following solutions to the flapping equation in forward flight.

$$\beta_{1s} - \theta_{1c} = \frac{-\frac{4}{3}\mu\beta_0}{1 + \frac{1}{2}\mu^2}$$

$$\beta_{1c} + \theta_{1s} = \frac{-\frac{8}{3}\mu \left(\theta_0 - \frac{3}{4}\lambda + \frac{3}{4}\mu\theta_{1s} \right)}{1 - (1/2)\mu^2}$$

Express flapping angle by Fourier expansion, take 1st harmonic term:

$$\beta = \beta_0 + \beta_{1c} \cos \psi + \beta_{1s} \sin \psi$$

$$\therefore \dot{\beta} = -\beta_{1c} \sin \psi + \beta_{1s} \cos \psi$$

$$\therefore \ddot{\beta} = -\beta_{1c} \cos \psi - \beta_{1s} \sin \psi$$

same for pitching angle, $\theta = \theta_0 + \theta_{1c} \cos \psi + \theta_{1s} \sin \psi$

Find the lateral input response by setting $\theta = \theta_{1c} \cos \psi$ and using harmonic matching. β_0 will not be affected as shown above, but β_{1s} and β_{1c} will

For $\theta_{1c} = 5^\circ$, $\mu = 0.4$ and $\beta = 2^\circ$

$$\beta_{1s} = 4.01^\circ$$

the TPP will have cyclic change by $4.01^\circ \sin \psi$